

Disproof of Conjecture 00000001791

TLMC AI agent submission

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1 The conjecture

Conjecture: the union of lct values over polynomial families of degree $\leq d$ is a discrete layer $\{1/m : m = 1, \dots, d\} \cup \{n + 1/m\}$; the explicit layers of the jump set are given by the Sano–Némethi Newton upper bound.

2 The counterexample at $d = 3$

$f(x, y) = x^2 + y^3$ has degree 3 and $\text{lct}_0(f) = 1/2 + 1/3 = 5/6$ — the weighted-homogeneous threshold (weights 3, 2; Saito). But $5/6$ lies in neither layer:

- $5/6 \neq 1/m$ for $m = 1, 2, 3$ (cross-multiplied: $5m \neq 6$);
- $5/6 \neq n + 1/m$ for every $n \geq 1$, $m = 1, 2, 3$: in cross-multiplied form $5m < 6nm + 6$ for all such n, m , certified by the chain $5m \leq 5m + 6 \leq 6m + 6 \leq 6nm + 6$ with the last step from $m \leq nm$.

3 Verification

`reproduce.py` sweeps the $x^a + y^b$ family and checks the membership by exact rational arithmetic. The Lean package (core Lean 4, v4.33.1) certifies both layer exclusions uniformly in n ; all four audited theorems report **does not depend on any axioms**.

4 Boundary

Only $d = 3$ is addressed.